

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026 CHEMISTRY – SSC-I (9th Grade)

Total Marks: 65

Time Allowed: 2 Hours 30 Minutes

Version: 1

General Instructions:

- Section A** (MCQs): Attempt **ALL 12** questions. Each carries **1 mark**. Time: 15 minutes.
- Section B** (Short Questions): Attempt **11**, choosing main **OR** alternative. Each **3 marks**.
- Section C** (Long Questions): Attempt **4**, choosing main **OR** alternative. Each **5 marks**.
- Use black or blue ink only. Pencil is not allowed except for diagrams.

SECTION A – Multiple Choice Questions (12 × 1 = 12 Marks)

Encircle the correct option. All questions are compulsory.

#	Question	A	B	C	D	Answer
1	The subatomic particle that has no charge is:	Proton	Electron	Neutron	Positron	(A)(B)(C)(D)
2	The atomic number of an element is equal to the number of:	Neutrons	Protons	Nucleons	Shells	(A)(B)(C)(D)
3	The number of neutrons in $^{37}_{17}\text{Cl}$ is:	17	18	20	37	(A)(B)(C)(D)
4	The maximum number of electrons that a p sub-shell can accommodate is:	2	6	10	14	(A)(B)(C)(D)
5	Group 1 elements of the periodic table are called:	Halogens	Noble gases	Alkali metals	Chalcogens	(A)(B)(C)(D)
6	The total number of periods in the modern periodic table is:	5	6	7	8	(A)(B)(C)(D)
7	The element with the highest electronegativity in the periodic table is:	Oxygen	Fluorine	Chlorine	Nitrogen	(A)(B)(C)(D)
8	As we move from left to right across a period, the atomic size:	Increases	Decreases	Remains same	First increases then decreases	(A)(B)(C)(D)
9	A chemical bond formed by the complete transfer of electrons is called:	Covalent bond	Ionic bond	Metallic bond	Hydrogen bond	(A)(B)(C)(D)
10	The number of shared pairs of electrons in a nitrogen molecule (N_2) is:	One	Two	Three	Four	(A)(B)(C)(D)
11	Which of the following compounds contains a coordinate covalent bond?	NaCl	H_2O	NH_4^+	CH_4	(A)(B)(C)(D)
12	Hydrogen bonding occurs when hydrogen is bonded to:	C, N, S	N, O, F	Cl, Br, I	Na, K, Li	(A)(B)(C)(D)

SECTION B – Short Questions (11 × 3 = 33 Marks)

Attempt 11 questions. Choose either the main question OR its alternative.

Q.	Question	Marks		OR Question	Marks
2(i)	State the main postulates of Dalton's atomic theory. Why is it considered the foundation of modern chemistry?	3	OR	State any three defects in Rutherford's atomic model.	3
2(ii)	Compare the relative charge and relative mass of proton, neutron, and electron in tabular form.	3	OR	Differentiate between atomic number (Z) and mass number (A) with one example.	3
2(iii)	Define isotopes. Describe the three isotopes of hydrogen with their symbols and number of neutrons.	3	OR	State any three uses of radioactive isotopes in medicine and research.	3
2(iv)	Differentiate between a cation and an anion with one example of each.	3	OR	Write the electronic configuration of $_{11}\text{Na}$, $_{13}\text{Al}$, and $_{17}\text{Cl}$.	3
2(v)	Define periodic table and state the modern periodic law.	3	OR	Differentiate between a group and a period in the periodic table.	3
2(vi)	Define shielding effect. How does it vary across a period and down a group?	3	OR	Define ionization energy. How does it change across a period and down a group?	3
2(vii)	State any three characteristic properties of alkali metals (Group 1 elements).	3	OR	Why are noble gases (Group 18) chemically inert? Explain in terms of electronic configuration.	3

2(viii)	Compare the general physical properties of metals and non-metals (any three differences).	3	OR	Describe any three characteristic properties of transition elements.	3
2(ix)	State the octet and duplet rules. Give one example of each.	3	OR	Why do atoms react with each other? Explain with reference to noble gas configuration.	3
2(x)	Define ionic bond. Show the formation of NaCl using electron dot and cross structures.	3	OR	Differentiate between polar and non-polar covalent bonds with one example of each.	3
2(xi)	Define coordinate covalent bond. Show the formation of ammonium ion (NH_4^+) as an example.	3	OR	What is hydrogen bonding? Explain with the example of water molecules.	3

SECTION C — Long Questions ($4 \times 5 = 20$ Marks)

Attempt 4 questions. Choose either the main question OR its alternative.

Q.	Question	Marks		OR Question	Marks
3(i)	State the main postulates of Bohr's atomic theory. How does Bohr's model overcome the defects of Rutherford's model?	5	OR	Describe Rutherford's gold foil experiment in detail. State the conclusions he drew and the defects of his planetary model.	5
3(ii)	Define electronic configuration. Explain the concept of shells, sub-shells, and the Aufbau principle with examples.	5	OR	Define isotopes. Describe the isotopes of hydrogen, carbon, and chlorine with their natural abundance and uses.	5
3(iii)	Explain the periodic trends in atomic size, ionization energy, electron affinity, and electronegativity across a period and down a group.	5	OR	Describe the halogens (Group 17) in detail including their appearance, electronic configuration, reactivity, displacement reactions, and thermal stability of hydrogen halides.	5
3(iv)	Explain the formation of a covalent bond. Describe single, double, and triple covalent bonds with the examples of H_2 , O_2 , and N_2 using electron dot and cross diagrams.	5	OR	Define metallic bond. Explain the electron-sea model and use it to account for the malleability, ductility, and electrical conductivity of metals.	5

Invigilator's
Signature _____

Candidate's
Signature _____

Chapters Covered: Ch 3 — Atomic Structure | Ch 4 — Periodic Table and Periodicity of Properties | Ch 5 — Chemical Bonding
— End of Question Paper —